

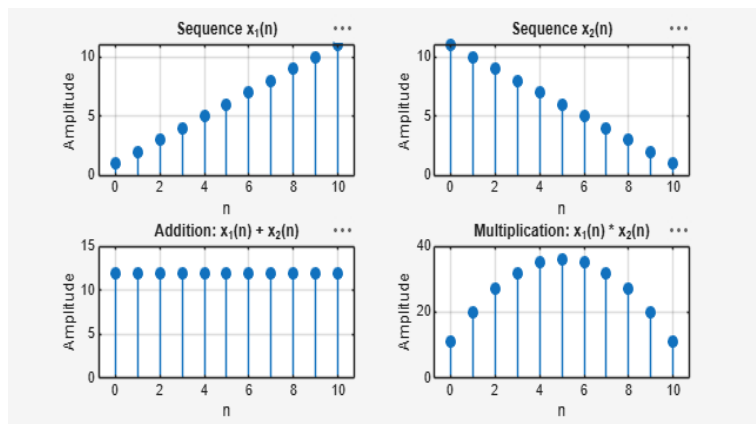
EXPERIMENT 3

SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

3.1. Addition and Multiplication of Two Sequences

```
% Addition and Multiplication
of Two Sequences
clc;
clear;
close all;
% Sample index
n = 0:10;
% Sequence 1
x1 = [1 2 3 4 5 6 7 8 9 10
11];
% Sequence 2
x2 = [11 10 9 8 7 6 5 4 3 2
1];
% Addition of sequences
y_add = x1 + x2;
% Multiplication of sequences
y_mul = x1 .* x2;
% Plotting
figure;
subplot(2,2,1);
stem(n,x1,'filled');
grid on;
title('Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
stem(n,x2,'filled');
grid on;
title('Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,3);
stem(n,y_add,'filled');
grid on;
title('Addition: x_1(n) +
x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,y_mul,'filled');
grid on;
title('Multiplication: x_1(n)
* x_2(n)');
xlabel('n');
ylabel('Amplitude');
```

OUTPUT



3.2. Energy and Power Calculation of Given Signals

```
clc;
clear;
close all;

% Time index
n = -14:14;

% Signal
x = [zeros(1,10) 1 2 3 4 5 4 3 2 1
zeros(1,10)];

% Energy
E = sum(abs(x).^2);

% Power
P = E/length(x);

% Display
disp(['Energy of the signal = ',
num2str(E)]);
disp(['Power of the signal = ',
num2str(P)]);

% Plot
figure;
stem(n,x,'filled');
grid on;
xlabel('n');
ylabel('Amplitude');
title('Discrete-Time Signal');
```

OUTPUT

